

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic · 360 trials
Total training paths	1 · 360 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports/RandomForest

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	Flex
Sensor columns	20 channels
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	—	—	—
left thumb	✗	✗	✓	✓
left index	✗	✗	✓	✓
left middle	✗	✗	✓	✓
left ring	✗	✗	✓	✓
left pinky	✗	✗	✓	✓
left wrist	✗	—	—	—
right palm	✗	—	—	—
right thumb	✗	✗	✓	✓
right index	✗	✗	✓	✓
right middle	✗	✗	✓	✓
right ring	✗	✗	✓	✓
right pinky	✗	✗	✓	✓
right wrist	✗	—	—	—

Enabled: 20 · Disabled: 24 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 1080
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=1440 · test=72

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-* = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Pre c	W-Rec	W-F1
Random Forest	1.0000	0.0000	1.0000	0.0000	0.9861	0.9872	0.9861	0.9861	0.9872	0.9861	0.9861

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_ClosedHand_Flip_OpenHand	1.00	1.00	1.00	12
Double_FingerTips	1.00	1.00	1.00	12
Double_Okay_Ring_Middle_Pinky	1.00	1.00	1.00	12
Double_OpenHand_Flip_ClosedHand	0.92	1.00	0.96	12
Double_OpenHand_LPunch_R_Punch	1.00	0.92	0.96	12
Double_Snap	1.00	1.00	1.00	12
accuracy			0.99	72

macro avg	0.99	0.99	0.99	72
weighted avg	0.99	0.99	0.99	72

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.5833

Per-class external accuracy

Class	Random Forest	Support
Double_ClosedHand_Flip_OpenHand	0.333	6
Double_FingerTips	0.667	6
Double_Okay_Ring_Middle_Pinky	0.000	6
Double_OpenHand_Flip_ClosedHand	0.667	6
Double_OpenHand_LPunch_R_Punch	0.833	6
Double_Snap	1.000	6